

Mahendra Ranwa

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SUMMARY

Computer Engineering Technology student with hands-on experience in embedded systems, microcontrollers, and real-time applications. Skilled in C programming, STM32, PIC24, and hardware-software integration

SKILLS

- Programming: C, Java, Python
- Embedded Systems: STM32, PIC24, Arduino
- Tools: MPLAB X, STM32CubeMX, Git, Anaconda, AutoCAD
- Hardware: Circuit design, debugging, sensors, ADC/DAC, PCB soldering
- Systems: Linux/UNIX, Networking basics
- Standards: IPC-A-610, IPC J-STD-001

EDUCATION

Computer Engineering Technology (Advanced Diploma)
Sheridan College

PROJECTS

Digital Multimeter (DMM) Fabrication | Electrical Fabrication

- Assembled and soldered a functional digital multimeter circuit
- Performed PCB-level soldering following electronic assembly practices
- Tested and verified voltage, current, and resistance measurements
- Diagnosed and corrected circuit faults during assembly
- Ensured accurate functionality through validation and troubleshooting

Embedded Control and Signal Processing System | C, STM32 (Nucleo L476)

- Developed embedded applications using GPIO, timers, ADC, interrupts, PWM, and input capture
- Designed timer-based systems (TIM6, TIM7) for precise delays using PSC and ARR calculations
- Implemented interrupt-driven control with EXTI and timer interrupts for real-time response
- Generated and analyzed PWM signals and measured signal characteristics using input capture
- Integrated ADC and analog watchdog for real-time monitoring and system control

Embedded Systems Application (ADC, Timers, Interrupts, UART) | C, PIC24 (Explorer 16)

- Built multi-module system integrating ADC, timers, interrupts, GPIO, and UART communication
- Acquired and processed real-time sensor data with interrupt-driven ADC
- Transmitted formatted data via UART (9600 baud) to terminal interface

- Implemented LED control logic with variable timing and pattern generation
- Designed interrupt-based system for timing, control, and user interaction

Smart Car System (3-Wheel Buggy) | Raspberry Pi, Python

- Built and assembled a 3-wheel smart car integrating motors, sensors, camera, and control modules
- Configured Raspberry Pi GPIO for motor control, sensor interfacing, and peripheral communication
- Implemented remote control and video streaming for real-time monitoring
- Integrated ultrasonic sensor for distance measurement and obstacle awareness
- Developed modular control logic for movement, sensing, and system coordination

EXPERIENCE

Web Technician (Volunteer) | U+ Toastmasters Academy

Sep 2025 – Mar 2026

- Managed website updates and digital content for community platform
- Maintained system reliability and supported technical operations
- Collaborated with teams to improve digital communication workflows

Support Technician | Sheridan College IT Centre, Brampton

Apr 2024 – Aug 2024

- Provided technical support for hardware, software, and network issues
- Installed and configured lab systems and academic software
- Diagnosed and resolved system faults to minimize downtime
- Assisted users with OS, connectivity, and application troubleshooting

Tutor (Electronics Fabrication) | Sheridan College

May 2023 – Dec 2023

- Assisted students with soldering techniques, PCB assembly, and component placement
- Guided troubleshooting of circuit faults and measurement using lab equipment
- Explained electronic fabrication concepts and safe handling practices
- Supported hands-on lab sessions to improve practical skills and accuracy

CERTIFICATIONS

IT Service Desk: Service Management

- Issued: March 2026
- Focus: IT service management, support workflows, and ticket handling

Computer Components and Peripherals for IT Technicians

- Issued: March 2026
- Focus: hardware components, system maintenance, and technical support